

Department of Earth System Sciences
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BIOGRAPHY

I am a physical oceanographer who uses ocean observations to investigate ocean dynamics and circulation in a changing climate. I have a particular interest in problems spanning scales (from turbulence to the large-scale overturning circulation) or spheres (e.g., biogeosphere), and in methods that leverage traditional observations with new platforms and satellite data.

PROFESSIONAL EXPERIENCE

Professor , Universität Hamburg, Germany	2022–present
Scientific leader , German Research Fleet Coordination Centre	2024–present
Science Leader , National Oceanography Centre, UK	2022
Principal Research Scientist , National Oceanography Centre, UK	2018–22
Associate Professor , University of Southampton, UK	2016–2018
Visiting Scientist , NASA Jet Propulsion Laboratory, USA	2016
Lecturer , University of Southampton, UK	2012–16
Senior Research Fellow , National Oceanography Centre, UK	2009–12

EDUCATION

Ph.D. in Physical Oceanography , University of Washington (with Peter Rhines)	2009
M.Sc. in Applied Mathematics , University of Washington	2009
M.Sc. in Oceanography , University of Washington (with Eric Kunze)	2005
A.B. in Applied Mathematics , Harvard University (with Ana Barros)	2002

AWARDS & HONORS

Ocean Observing Team Award, The Oceanography Society	2021
Nicholas P. Fofonoff award, American Meteorological Society	2021
EGU Outstanding Early Career Scientist award	2017
Steinbach Scholar at Woods Hole Oceanographic Institution (WHOI)	2016
Vice Chancellor's teaching award (UoS FNES, £1000 prize)	2015
Fellow of the Higher Education Academy (FHEA)	2015
Excellence in Teaching Award, category: Best Feedback (UoS FNES)	2014
Outstanding student paper award, AGU/ASLO Ocean Sciences	2008
WHOI Geophysical Fluid Dynamics Fellowship	2004
UW Program on Climate Change Fellowship	2002
Summer Undergraduate Research Fellow at Scripps Inst. of Oceanogr.	2000
Certificate of Distinction for teaching (Harvard)	2000
Research Science Institute at the Massachusetts Institute of Technology	1997

FUNDING

DFG Large equipment, Ocean Array (PI), €1.04M	2026–27
EIG-CONCERT-Japan, UHEAT (co-I), €437k (of which €147k to UHH)	2026–29
DFG, PycnMix (PI), €585k	2025–28
AEI-DFG, DS MIXSED (PI), €932k (of which €466k to UHH)	2025–28
DFG Large equipment, Swarm of Gliders (PI), €2.2M	2024–26
EU Horizon Europe, EPOC (PI), €8M	2022–27
UKRI-funded (ERC) fellowship PycnoGen (co-I), €3.5M	2022–27

NERC Highlight topic, DEFIANT (co-I), £5M	2021–25
NERC UK BGC Argo array (co-I), £1.5M	2021–23
NERC Next generation multi-disciplinary array, BGC-RAPID (co-I), £570k	2021–22
NERC Net Zero Oceanographic Capability, NZOC (co-I), £250k	2020
NERC Large Grant, DeCAdeS (co-I), NE/T012714/1, £3.4M	2020–25
Lloyd's Register autonomy demonstrator, ALADDIN (co-I), £165k	2020
ERC Starting Grant Fellowship, TERIFIC (PI), 803140, €1,999k	2018–26
NERC Standard grant, BLT Recipes (co-I), NE/S001433/1, £889k	2018–23
NERC Standard grant, DynOPO (co-I), NE/K013181/1, £968k	2015–19
NERC Standard grant, MerMEED (PI), NE/N001745/1, £1,048k	2016–19
NERC Technology grant, FreshWATERS (co-I), NE/P003176/1, £171k	2016–17
NERC Sensors on AUVs, GliSENEx (co-I), NE/J020184/1, £150k	2013–17
Leverhulme Trust Research Fellowship (PI), £14k	2016
Southampton Marine & Maritime Institute stimulus fund (co-I), £15k	2016
Huckabay Teaching Fellowship , UW	2008
National Science Foundation Graduate Research Fellowship, 3 years	2004–07
National Defense Science & Engineering Graduate , Fellowship, 2 years	2002–04

FIELD EXPERIENCE (responsibility in brackets)

FS Meteor, M212 (to-yos), EPOC recovery, 5 weeks	Jul 2025
FS Maria S. Merian, MSM121 (pressure sensors), EPOC deployment, 3 weeks	Sep 2023
Qaqortoq, Greenland (gliders), Small boat, 2 weeks	Dec 2021
RRS Discovery (instrument allocations), RAPID moorings cruise, 8 weeks	Dec 2020
RRS James Cook (instrument allocations), RAPID moorings cruise, 4 weeks	Mar 2020
Qaqortoq, Greenland (team lead), Small boat, 1 week Aug, 2 weeks	Dec 2019
R/V Walton Smith (training the PSO), MerMEED VMP/ADCP cruise, 2 weeks	Mar 2018
R/V Walton Smith (as PSO), MerMEED VMP/ADCP cruise, 2 weeks	Oct 2017
RRS James Clark Ross (Autosub), DynOPO process cruise, 7.5 weeks	Mar 2017
R/V Walton Smith (as PSO), MerMEED VMP/ADCP cruise, 1 week	Dec 2016
RRS James Clark Ross (CTD), DynOPO moorings & A23 section, 5 weeks	Mar 2015
RRS James Cook (underway/ADCP), RAPID moorings cruise, 6 weeks	Apr 2014
R/V Knorr (as UK PSO), RAPID moorings cruise, 3 weeks	Apr 2011
RRS Discovery (moorings), RAPID moorings cruise, 5 weeks	Dec 2010
R/V Wecoma (CTD/XCP), Internal waves over the Oregon slope, 2 weeks	Sep 2005
R/V Wecoma (microstructure/XCP), Hawaiian ridge waves & mixing, 3 weeks	Aug 2002
R/V Revelle (CTD/radiosonde), Juan de Fuca ridge movement, 2 weeks	Aug 2000

TEACHING EXPERIENCE (UHH = Hamburg, UoS = Southampton)

63-713 ADVANCE: Data Analysis for Physical Oceanography, UHH	2026
63-770 Atlantic Meridional Overturning Circulation, UHH	2025
63-713 ADVANCE: Sea-going oceanography, UHH	2024
63-710 Oceanic Exercises, UHH	2024
63-716/7 Regional Oceanography, UHH	2023–24, 2026
Physics of the Ocean summer school, Bad Honnef, DE	2023
63-705/6 Observational Methods and Remote Sensing, UHH	2022–25
Proposal writing (5 sessions) for ECRs, NOC	2019
NEXUSS Statistics & Data Analysis, NOC	2018
ISNAO summer school, Bonne Bay, Canada	2017
SOES3010/6005: Large Scale Ocean Processes, UoS	2014–17
SOES2025: Methods in Oceanography, UoS	2014–17

SOES3018: Falmouth fieldwork course, UoS	2017
SOES6070: Advanced fieldwork course, UoS	2012–14
SOES3035: Research training, UoS	2013
SOES3016: Oceanography from Space, UoS	2012, 2013
OCN506: Communicating Science with Figures, UW	2008

MENTORSHIP AND SUPERVISION

Postdocs/Research Scientists: Felipe Vilela da Silva (co-, 2025–present), Joel Bracamontes Ramírez (2024–present), Simon Wett (2024–present), Elodie Duyck (2023–25), Louis Clément (2020–22), Darren Rayner (2020–22), Alejandra Sanchez-Franks (2019–21), Ben Moat (2019–22), Ilona Goszczko (2019–21), Carl Spingys (co-, 2017–20), D. Gwyn Evans (2016–19), Cristian Florindo-Lopez (2016)

PhD Students: Till Moritz (2026–present), Gema Camacho Viteri (panel, 2025–present), Lara Heyl (2024–present), Emelie Breunig (2023–present), Maria-Jesus Rapanague (panel), Heather Jacques (co-), Morag Forthingham (panel), Chris Auckland (co-, PhD'25), Markus Ritschel (co-, PhD'24), Manish Devana (committee, PhD'23), Delphine Lobelle (co-, PhD'19), Neela Morarji (co-, PhD'18), Freya Garry (PhD'17), Lena Schulze (PhD'16), Victoria Hemsley (co-, PhD'16), Louis Clement (PhD'14)

BSc and MSc dissertations: >30 dissertations supervised since 2010. Recent: Susie Brunier, Lea Dietz, Clemens Rohling, Leonie Bauer, Till Moritz, Isabelle Schmitz, Katja Schultz, Boris Shapkin, Viktoria Nikolaus, Julian Morales Meabe. Of note: Jemima Rama[†] (MSci, 2016), Jo Ribeiro[†] (MSci, 2015), Lisa Holton* (BSc, 2013), Maren Richter (Kiel Univ., 2014) and Atul Kumar Yadav (IIT Bhubaneswar, 2013). **dissertation award*, [†]*top student award*

PROFESSIONAL ACTIVITIES

Committees:

AMS Oceanographic Research Awards Committee	2024–26
UHH MIN Faculty: Large equipment	2023–present
CLIVAR AMOC Task Team, co-chair	2021–present
CLIVAR Atlantic Regional Panel (ARP) co-chair	2021–24
CLIVAR Atlantic Regional Panel (ARP) member	2019–20
Royal Society Newton International Fellowships, Physical Sciences	2018–22
NERC Peer Review College member	2016–22
NEXUSS Centre for Doctoral Training (co-Director)	2017–18
Women in Ocean and Earth Sciences at Southampton	2014–16
UoS Employability representative	2012–16
UW Student-faculty representative	2008

Organisation of Conferences / Workshops / Seminars:

AMOC BGC Workshop: BGC Sensors on AMOC Arrays	2026
AGU Ocean Sciences, Glasgow: Atlantic Connectivity	2026
CLIVAR AMOC Workshop: Observation needs in a changing climate	2023
NZOC Workshop: 21st century marine scientist	2021
EGU General Assembly, Vienna: Ocean Circulation	2019
IUGG general assembly, Prague: MOC & Deep Currents	2015
AGU Fall Meeting, San Francisco: AMOC, climate variability and change	2014
AGU Ocean Sciences, Honolulu: Frontiers in Oceanographic Data & Methods	2014
US AMOC/UK RAPID international meeting, Baltimore	2013
IAPSO meeting, Gothenburg: Thermohaline circulation and deep currents	2013
EGU General Assembly, Vienna: Ocean Circulation	2013

EGU General Assembly, Vienna: Ocean Circulation	2012
AGU Ocean Sciences, Salt Lake City: Vertical Flow in the Ocean	2012
NOC: Physical Oceanography and Climate Seminar	2010–11
UW: Student Physical Oceanography educational Retreat, Friday Harbor	2003, 2009
UW: Graduate Climate Change Conference, Pack Forest	2008

Other activities:

Maria de Maeztu International Visiting Scholar, Esportes, Spain	2025
RAPID/OSNAP Data Workshop, WHOI, USA	2025
International Space Science Institute (ISSI) international team, "Relationship between the AMOC and coastal sea level", Bern, Switzerland	2025–27

Outreach/Other Activities: Women in Oceanography, International Maritime Museum of Hamburg (2026), Hamburg Research Academy: Paving your Postdoc Career (2024), Scientific Career & Parenthood, panel member (2024), Royal Institution Christmas Lectures, guest on episode 2 (2020), RRS Sir David Attenborough launch, talk & marine robotics stand (3 days, 2019), Soapbox Science & Art presenter, Bournemouth Arts Festival (2018), Talked to 300 school kids from Springhill Primary (2018), Kid's version of "heat wave" paper (2016), Discover Oceanography on "Oceanography from Space" to U3A (2015), STEMnet ambassador, Hampshire (2014), Ocean and Earth Day demos for Science & Engineering week, NOC (2012, 2013, 2019)

Reviews for Journals: Annals of Glaciology, Deep Sea Research, Frontiers in Marine Science, Geophysical Research Letters, Journal of Atmospheric and Oceanic Technology, Journal of Climate, Journal of Geophysical Research - Oceans, Journal of Physical Oceanography, Marine Technology Society Journal, Nature, Nature Communications, Nature Geoscience, Ocean Science, Progress in Oceanography, Remote Sensing of the Environment, Reviews of Geophysics

Reviews for proposals: UK Natural Environment Research Council (NERC), US National Science Foundation (NSF), Royal Society International Fellowships, Norwegian Research Council, National Defense Science & Engineering Graduate (NDSEG) fellowship, NASA Earth & Space Science Fellowships (NESSF), German research vessels (GPF), EuroFleets vessels

TRAINING AND CERTIFICATION

Instruments & Autonomous Vehicles: Seabird instruments, Kempten, 2 days (11/2025), Seaglider pilot training incl. under ice, UW, 5 days (6/2025), Sailbuoy pilot training, Offshore Sensing AS, 5 days (6/2019), Seaglider pilot training, Kongsberg, 5 days (9/2017)

Safety & First aid: Deutsches Rotes Kreuz First Aid, 8 hours (11/2022), IOSH Managing Safety in a Research Environment, 15 hrs (10/2018), First Aid at Work, 15 hrs (11/2019), ITC Certificate in Outdoor First Aid, SCQF Level 5, 16 hrs (2/2015)

Seagoing: Certificate in Proficiency in Designated Security Duties, 10 hrs (9/2020); STCW Personal Survival techniques certificate (updated 1/2017, 2010); ENG1 seafarer medical fitness certificate (1/2020, 3/2014); British Antarctic Survey medical (9/2014)

Diving: PADI Advanced diver No. 0009962148 (2000), PADI Open Water (1996)

Teaching: "PhD Supervision, MIN Faculty, UHH", 9 hours (2024); "Flipped Learning", 4 hrs (2015); "Revitalising your Virtual Learning Environment", 2 hrs (2015); Postgraduate certificate in academic practice (PCAP) training, 24 hrs (2013); "Engaging Students in Research & Inquiry", 3 hrs (2013); "Effective Teaching and Learning in the Large Classroom Setting" by NAGT, 4 hrs (2012); "Supervising a PhD student," 3 hrs (2010)

Other: "Excelling at Academic Interviews," 7 hrs (2015); "Springboard: Women's development programme," 32 hrs (2015); "ThinkWrite: Quality Papers", 7 hrs (2013); "Building & Leading High Performing Teams", 7 hrs (2013); "Managing your Academic Career: for Women", 7 hrs (2013); "Climate Communications: Tools & Tips" at AGU fall mtg, 7 hrs (2012)

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- [70] Jacques*, Clément, **Frajka-Williams**, Naveira Garabato, and Oltmanns. "Thermal vs haline drivers of springtime restratification in the Labrador Sea". *J. Geophys. Res. Oceans* (2026). doi: 10.22541/essoar.176365845.54878695/v1.
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- [8] McCarthy, **Frajka-Williams**, Johns, Baringer, Meinen, Bryden, et al. "Observed interannual variability of the Atlantic MOC at 26.5°N". *Geophys. Res. Lett.* (2012). doi: 10.1029/2012GL052933.
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- [6] **Frajka-Williams**, Cunningham, Bryden, and King. "Variability of Antarctic Bottom Water at 24.5°N in the Atlantic". *J. Geophys. Res. Oceans* (2011). doi: 10.1029/2011JC007168.
- [5] **Frajka-Williams**, Eriksen, Rhines, and Harcourt. "Determining vertical water velocities from Seaglider". *J. Atmos. Oceanic Tech.* (2011). doi: 10.1175/2011JTECH0830.1.
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- [3] **Frajka-Williams** and Rhines. "Physical controls and interannual variability of the Labrador Sea spring phytoplankton bloom in distinct regions". *Deep-Sea Res. Pt. I* (2010). doi: 10.1016/j.dsr.2010.01.003.
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BOOK CHAPTERS AND REPORTS (selected)

- [10] **Frajka-Williams**, Brearley, Nash, and Whalen. "New technological frontiers in ocean mixing". *Ocean Mixing: Drivers, Mechanisms and Impacts*. Elsevier, 2022. doi: 10.1016/B978-0-12-821512-8.00021-9.

- [9] Hendry, Annett, Bhatia, Damerell, Fielding, Firing, et al. "Equity at sea: Gender and inclusivity in UK sea-going science". *Ocean Challenge* (2020). url: <https://nora.nerc.ac.uk/id/eprint/530066/>.
- [8] **Frajka-Williams**. "Topographic eddies". *Reference Module in Earth Systems and Environmental Sciences*. Elsevier, 2018. doi: 10.1016/B978-0-12-409548-9.10852-8.
- [7] **Frajka-Williams**. *RV Walton Smith Cruise WS17305, 31 Oct - 10 Nov 2017, Miami to Miami, USA. MerMEED microstructure cruise report*. Tech. rep. National Oceanography Centre, Southampton, 2018. url: <https://eprints.soton.ac.uk/417559/>.
- [6] **Frajka-Williams** and Templeton. *Boaty McBoatface M44 in Orkney Passage*. <https://vimeo.com/eleanorfracja/boaty-mcboatface-m44-dynopo>. Animation. Viewed 45,000 times. Vimeo, 2017.
- [5] **Frajka-Williams**. *RV Walton Smith Cruise WS16336, 01 - 07 Dec 2016, Miami to Miami, USA. MerMEED microstructure cruise report*. Tech. rep. National Oceanography Centre, Southampton, 2017. url: <https://eprints.soton.ac.uk/410888/>.
- [4] Duchez, **Frajka-Williams**, Josey, Evans, Grist, Marsh, et al. "Cold ocean = hot summer?" *Environmental Science Journal for Teens* (2016). url: <http://www.sciencejournalforkids.org/articles/cold-ocean-hot-summer>.
- [3] **Frajka-Williams**. "Women in Oceanography: A decade later". *Oceanogr.* (2014). url: <https://tos.org/oceanography/issue/volume-27-issue-04-supplement>.
- [2] Johns and **Frajka-Williams**. *RV Knorr Cruise KN200-4, 13 Apr-3 May 2011. RAPID Mooring Cruise*. Tech. rep. National Oceanography Centre, Southampton, 2011. url: https://nora.nerc.ac.uk/id/eprint/308915/1/NOC_CR_07.pdf.
- [1] Martini, **Frajka-Williams**, and Mouw. "Conference Report | The Pattullo Conference: Building community through mentoring". *Oceanogr.* (2009). doi: 10.5670/oceanog.2009.26.

SELECTED SEMINARS & TALKS (as presenter)

- 2025: Klima Campus Colloquium, Hamburg, Germany (seminar)
 GEOMAR OCDD, Kiel, Germany (seminar)
 CELLO Conference, Hamburg, Germany (**invited talk**)
 IMEDEA, Esporles, Spain (seminar)
- 2024: ISMS meeting, Valencia, Spain (**keynote**)
 Ocean Sciences meeting, New Orleans, LA (talk)
 EUMETSAT Winter Talk, Darmstadt (seminar - online)
- 2023: AMOC workshop, Hamburg, Germany (talk)
 Universität Bremen, Germany (seminar)
 GEOMAR, Kiel, Germany (seminar)
 Bottom pressure workshop, Rhode Island (talk - online)
 ASOF meeting, Canary Islands (talk - online)
- 2022: AMOC meeting, Royal Society, London, UK (**invited talk**)
 AANChOR AAORIA Workshop, Washington D.C., USA (talk)
 UG2 Glider workshop, Seattle, USA (poster)
- 2021: Leeds, UK (seminar)
 NOC Science & Technology Advisory Committee, UK (talk)
 FDSE summer school, Cambridge, UK (lecture)
 Nordic Overflows workshop, virtual (talk)
 CANAIMOC meeting, virtual (talk)
 EGU General Assembly, virtual (pico)

- 2020: OceanSITES, virtual (**invited panelist**)
 NOC Board, UK (talk)
 IOCAG, Canary Islands, Spain (seminar)
 Oxford University, UK (seminar)
 UK MetOffice, UK (talk)
 Imperial College London, UK (seminar)
- 2019: Marine Autonomy & Technology Showcase, Southampton, UK (talk)
 GFDL, Princeton, USA (seminar)
 Newcastle University, Newcastle, UK (seminar)
 RRS Sir David Attenborough launch, Birkenhead, UK (talk)
 OceanObs19, Honolulu, Hawaii (poster)
 AMOC Metrics, Honolulu, Hawaii (**invited talk**)
 NERC Science Committee, Swindon, UK (talk)
 NOC Association, London, UK (talk)
 CLASS annual science meeting, Plymouth, UK (talk)
 EGU General Assembly, Vienna, Austria (poster)
 Royal Society West Indies meeting, Chicheley, UK (poster)
 RAPID International Review, London, UK (talk)
- 2018: Marine Autonomy & Technology Showcase, Southampton, UK (talk)
 University College London, London, UK (seminar)
 Challenger Society for Marine Science, Newcastle, UK (talk)
 US AMOC/UK RAPID International Meeting, Miami, FL (**invited talk**)
 University of East Anglia, Norwich, UK (seminar)
 Ocean Sciences meeting, Portland, OR (talk)
 Cambridge University, Cambridge, UK (seminar)
- 2017: Marine Autonomy & Technology Showcase, Southampton, UK (talk)
 RAPID/OSNAP/ACSIS meeting, Oxford, UK (poster)
 Oceans and Climate public lecture, The Royal Society, London (**keynote**)
 IAPSO meeting, Cape Town, South Africa (talk)
 Liege Colloquium on Turbulence, Liege, Belgium (poster)
 NOC Friday Seminar, Southampton, UK (seminar)
- 2016: EGO Glider meeting, Southampton, UK (poster)
 NOAA/AOML, Miami, FL (seminar)
 Woods Hole Oceanographic Institute, Woods Hole, MA (seminars)
 NASA JPL, Pasadena, CA (seminar)
 University of Washington, Seattle, WA (seminar)
- 2015: RAPID International Science Meeting, Bristol, UK (talk)
 IUGG General Assembly, Prague, Czech Republic (talk & **panel member**)
 CLIVAR Climate Process Team meeting, La Jolla, CA
 University of Washington, Seattle, WA (seminar)
- 2014: AGU fall meeting, San Francisco, CA (talk)
 Ocean Sciences, Honolulu, HI (talk)
 National Oceanography Centre, Liverpool, UK (seminar)
 Oxford University, Oxford, UK (seminar)
- 2013: IAPSO meeting, Gothenberg, Sweden (talk)
 Challenger Society: Prospectus 2013, Royal Society, London (**invited talk**)
 EGU General Assembly, Vienna, Austria (talk)
 University of Washington, Seattle, WA (seminar)
 University of East Anglia, Norwich, UK (seminar)

- 2012: AGU Fall Meeting, San Francisco, CA (poster)
 Bangor University, Bangor, UK (seminar)
 THOR meeting in Hamburg, DE (talk)
 British Antarctic Survey, Cambridge, UK (seminar)
 Time series conference in Brest, France (**invited talk**)
 USAMOC meeting, Boulder, CO (poster)
 EGU General Assembly, Vienna, Austria (talk)
 AGU Ocean Sciences, Salt Lake City, UT (poster)
- 2011: WCRP meeting, Denver, CO (poster)
 RAPID International Science Meeting, Bristol, UK (talk)
 ZMAW/Klimacampus, Max-Planck-Institut für Meteorologie, Hamburg (seminar)
 IUGG General Assembly, Melbourne, Australia (talk)
 IUGG General Assembly, Melbourne, Australia (poster)
- 2010: Challenger Society for Marine Science, Southampton, UK (poster)
 AGU Ocean Sciences, Portland, OR (talk)
 Imperial College London, London, UK (seminar)
 University of Liverpool, Liverpool, UK (seminar)
 POETS NOC, Southampton, UK (seminar)
 NOC PO Seminar, Southampton, UK (seminar)
- 2009: ESSAS 2009 Annual Science meeting, Seattle, WA (**invited talk**)
 PO and Climate, Southampton, UK (seminar)
 University of Washington, Seattle, WA (seminar)
 Woods Hole Oceanographic Institution, Woods Hole, MA (seminar)
 Physical Oceanography Dissertation Symposium. Honolulu, HI (talk)
- 2008: Ocean Sciences meeting, Orlando, FL (**Outstanding Student Talk award**)
 MPOWIR Pattullo Conference, Charleston, SC (talk)
- 2006: Ocean Sciences meeting, Honolulu, HI (poster)
- 2005: EGU General Assembly, Vienna, Austria (poster)
- 2004: American Physical Society, Seattle, WA (talk)
 SCOR IAPSO conference on Mixing, Victoria, Canada (poster)
 AGU Ocean Sciences, Portland, OR (poster)
- 2003: Hawaiian Ocean Mixing Experiment workshop, Mt. Hood, OR (talk)
- 2002: EGU General Assembly, Nice, France (poster)